Simon Vudiya Lirova

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Profile

I am a self-driven and passionate agricultural scientist committed to making society a better place through science-based solutions in agriculture. My goal is to bridge the gap between farming practices and cutting-edge research to improve productivity, ensure food safety, and promote the sustainable use of natural resources. I believe agriculture is central to driving sub-Saharan Africa toward greater development, industrialization, and meaningful employment for its youthful population. Beyond my vision, I bring practical expertise in molecular biology and biotechnology gained through my work in molecular laboratories, including the Kenya Plant Health Inspectorate Service (KEPHIS). I have hands-on experience in

DNA extraction using both CTAB buffers and commercial kits (e.g., Zymo Tech).

Running and interpreting gel electrophoresis for DNA analysis. Polymerase Chain Reaction (PCR) and post-amplification documentation for pathogen identification. Working with plant viruses (e.g., Banana Bunchy Top Virus, Begomoviruses) and bacterial pathogens (e.g., *Ralstonia solanacearum* and *Dickeya spp*.).

Using Loop-Mediated Isothermal Amplification (LAMP) for rapid diagnostics.

Primer reconstitution, equipment calibration, and strict adherence to laboratory protocols.

I am passionate about advancing molecular techniques and localizing their application within Africa's agricultural sector ensuring they are not only accessible to multinationals but also embedded in national systems for crop protection, disease management, and food security. I see myself not just as a researcher, but also as a team builder, leader, and innovator, inspiring peers to align their efforts toward the shared vision of a food-secure and self-reliant region.

Skills

Laboratory and Molecular Techniques

- DNA/RNA/total nucleic acid extraction and purity determination (CTAB method and commercial kits).
- Gel electrophoresis, PCR amplification, and documentation of results.
- Loop-Mediated Isothermal Amplification (LAMP) diagnostics.
- Primer reconstitution, preparation of stock solutions, and calibration of lab equipment.

- Sample preparation, accurate record-keeping, and adherence to laboratory protocols.

Field and Research Skills

- Designing and conducting field experiments, including layout and setup.
- Data collection on pests and diseases using KoboToolbox and ODK.
- Questionnaire development (Excel-based) for digital survey tools.
- Field mapping, survey execution, and data entry/cleaning.

Data Analysis and ICT

- Statistical analysis using SPSS, GenStat, R, and Python.
- Data visualization and report writing.
- Skilled in Microsoft Office (Word, Excel, PowerPoint) and poster design.

Organizational and Administrative Skills

- Scheduling activities and coordinating teams.
- Formal communication, report writing, and professional correspondence.
- Organizing colleagues and contributing to collaborative projects.

Academic background

I have completed a **master's** degree in **Plant Pathology** from the University of Nairobi. The program introduced me to logic reasoning, research work and matters on data analysis. I was able to master proposal writing skills and preparing scientific papers. I involved in collecting and collating data from surveys, collecting data from field and laboratory experiments, cleaning the data and analyzing to draw inferences from the sample to the original population. I am able to identify problem as it appears in the general population and come up with strategy to confirm before laying strategy to address.

I did my **Bachelors** degree of **Horticulture** from the University of Nairobi which I completed in 2018. Here I got insight on the importance of using modern farming techniques in addition to scientific innovation to address the problems experienced in agriculture. Most importantly is the idea of maintaining our environment for posterity. Horticulture opened my eyes to exterior landscaping and internal landscaping through ornamental plants and their therapeutic effect on humanity.

I completed my secondary education in 2012 where I did well to secure government scholarship to the competitive university programs. In my elementary and primary

education, I interacted with my immediate environment and friends which shaped my career after I realized the insufficient food that affected most of the homesteads.

Work experience

2025-Present -Kenya National Academy of Sciences (KNAS) – Program Officer

- Delegate duties and coordinate colleagues to ensure smooth operations.
- Plan and organize institutional events and meetings.
- Record and prepare minutes during high-level council meetings.
- Represent the Academy at conferences and professional forums.
- Manage and update organizational websites and databases.
- Conduct research and prepare proposals, reports, and official communications.
- Serve as steward of institutional property and resources.

2024 Kenya Plant Health Inspectorate service

Activities

i. Identification and classification of parasitic nematodes in Nematology laboratory.
ii. ii. Extract, denature and purify DNA/RNA from different sample types e.g. Nematodes fungal,
iii. Bacterial cultures, and plant material.
iv. Detection of plant pests using advanced molecular techniques such as Loop Mediated Isothermal Amplification (LAMP) and PCR (Real Time and Conventional)
v. Micro propagation of plant germplasm and virus clean up by tissue culture, meristem tip
vi. culture and thermotherapy.
vii. vii. Preparation of reagents, solutions and buffers required for molecular analysis.
viii. Performing routine testing, quality control, maintenance and proficiency testing in accordance with current laboratory procedures.
ix. General principles of Good Laboratory Practices and Quality Management Systems (under the ISO 17025 and 9001-2015 systems).
x. Report writing and sample reception using LIMs (Laboratory Information Management Systems) xi. General principles of Pest Risk Analysis (PRA) and pest surveillance. xii. Biosafety regulation and testing of Genetically Modified Organisms (GMOs)

2023- Agriculture and Food Authority

Activities

i. I served in the Technical Advisory Services of the Food crop Directorate where I collected data, cleaned and analyzed them producing reports.
ii. I also educated the Sub-County Agricultural Officers of the regulations in the Crop Acts 2019.
iii. iii. I organized stands in a number of Shows in Bukura, KALRO headquarters and ASK Shows.
iv. I prepared demonstration plots for alternative food crops with the Agricultural Training Centers in Kisii, Migori, Kisumu and Elgeiyo Marakwet.

2022- Centre for Agriculture and Biosciences International

Activities

i. Collecting data through surveys in all sub counties of Embu on fall army worm
ii. Collecting data on the adoption of mobile technology by farmers in Njoro, Molo, Kuresoi Noth and Kuresoi South

2021- Keya Agriculture and Livestock Research Organisation

- Carried out field research Experiments at Kalro Mwea where I screened 186 rice germplasm for their tolerance to brown spot, rice blast, and bacterial wilt at Kirogo field station

2018- Wildfire Horticulture

i. Hypericum and roses production from planting to harvesting
ii. Crop protection and management
iii. Crop nutrition and fertigation
iv. Crop husbandry and cultural practices
v. Quality Check and monitoring by scouting
vi. vi. Post-harvest treatment and grading of roses and hypericum
vii. vii. Health and safety adherence and provisions
viii. Employee management and work plan operations
ix. Produce of detailed report on work areas.
x. Supervision of Hypericum propagation as well as training in tissue culture.

2016- Oserian Development Company

i. Priva/water/irrigation system; including source of water, fertilizers classification and formulation, priva operations, and filter units operations
ii. Fertigation: Fertigation process and vermiculture process

iii. Supervision in propagation of statice: using the tissue culture method.

iv. Crop protection as well as training casuals; Identification of diseases and their casual agents – Powdery Mildew, Botrytis, Rust, Crown gall and their control. Identification of pests and their control – whiteflies, thrips, mites, caterpillar, aphids, mealy bugs, leaf miner. Management of soil-borne pests and diseases-nematodes, fusarium

2015- Finlay's Horticulture

i. The use of biopesticides to manage pests and diseases on horticultural crop

ii. How to rear the biopesticides(Ambysius californicus, Phytoseiulus persimilis

Trichoderma harzianum)

iii. Agronomic activities on the production of herbs and spices (Persely, mint, sage, tarragon, rosemary) and other crops including baby corn, sugar snaps, French beans, broccoli among others

REFEREES

1. Prof James Muthomi,

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